MIL–L–3077
31 October 1949
SUPERSEDING
Army 52–7–2D
14 February 1942

MILITARY SPECIFICATION

LINKS, METALLIC BELT, FOR SMALL ARMS AMMUNITION

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments

1. SCOPE AND CLASSIFICATION

1.1 Scope.—All links covered by this specification are intended to hold ammunition, and to form flexible belts.

1.2 Classification.

1.2.1. *Types.*—Metallic belt links shall be of the following types, as specified (see 6.1):

Link, metallic belt, caliber .30, M1.
Link, metallic belt, caliber .50, M2.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 Specifications.—The following specifications, of the issue in effect on the date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

QQ–M–151—Metals; General Specification for Inspection of

U. S. ARMY SPECIFICATIONS

52–0–1—General Specification Governing the Manufacture and Inspection of Small Arms Weapons, Spare and Replacement Parts, and Accessories.

57–0–2—Finishes, Protective, for Iron and Steel Parts.

57–136–1—Steel, Strip, for Metallic Belt Links.

94–40645—Marking; Exterior, Domestic and Export Shipment by Contractors.[1]

100–2—Standard Specification for Marking Shipments by Contractors.[2]

U. S. ARMY ORDNANCE DEPARTMENT TENTATIVE SPECIFICATION

AXS–1759—Compound, Rust Preventive, Soft Film.

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal and Military specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Marine Corps.*—Copies of Federal and Military specifications may be obtained upon application to the Quartermaster General, Headquarters, U. S. Marine Corps, Navy Building, Washington 25, D. C., or the Depot Quartermaster, Marine Corps Depot of Supplies, 1100 South Broad Street, Philadelphia 46, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Federal and Military specifications and U. S. Army Specification 94–40645 may be obtained upon application to the Commanding General, Air Materiél Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.)

2.3 Drawings.—The following drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY ORDNANCE DEPARTMENT DRAWINGS

C59722—Link, Metallic Belt, Cal. .30, M1.
D37987—Link, Metallic Belt, Cal. .50, M2.

[1] Applicable only to Air Force shipments.
[2] Applicable only to Army shipments.
[3] Applicable only to Navy shipments.

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(Copies of Army Ordnance Department drawings may be obtained upon application to the Chief of Ordnance, Department of the Army, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Components.

3.1.1 *General.*—Unless otherwise specified, links shall conform to specification 52–0–1.

3.1.2 *Material.*—Unless otherwise specified, the contractor shall use steel of such composition, form, and heat treatment as he considers best suited to produce links conforming to this specification. Prior to production the contractor shall furnish the contracting agency with a copy of complete specifications covering the type of steel to be used (see 6.2).

3.2 Physical requirements.

3.2.1 Links shall be free of laminated metal and shall conform to the dimensions shown on the applicable drawing. They shall be free of fractures and forming tears which would affect the functioning of the link. The links shall be free of burrs which would affect functioning of the gun, scratch the cartridge case or feeding components, or injure personnel while handling. Link defects shall be visually examined and gaged for the defects shown in table IA or table IB, as applicable.

3.2.2 *Pull strength.*—The contractor shall subject each link to the load required by the drawing. This requirement applies only to the caliber .30, M1, and caliber .50, M2, links. Sample links which are subjected by the Government inspector to the load required by the drawing shall not show an increase in center distance of more than 0.005 inch.

3.2.3 *Extraction strength.*—The extraction strength of each link shall be within the limits shown on the applicable drawing.

3.2.4 *Contractor's processing.*—The contractor shall determine and maintain a system of inspection including the necessary gages and equipment to insure that links conform to the physical requirements of this specification.

3.2.5 *Acceptable quality levels.*—Acceptable quality levels (percent defective as determined in accordance with table II or table III whichever is used) of the links with respect to visual, dimensional, and strength defects shall not exceed 3 percent for defective links containing either major or minor defects (see 6.5).

TABLE IA.—*Classification of defects for caliber .30 links*

Column I Point of inspection—gaging, functioning, and visual	Column II Limits of incidental defects	Column III Limits of minor defects	Column IV Limits of major defects
Connecting loop, inside diameter.	Basic dimension: +0.012. −0.002.	Basic dimension: +0.014. −0.003.	Exceeding tolerances of column III.
Neck loop, inside diameter.	Basic dimension: +0.010. −0.002.	Basic dimension: +0.012. −0.003.	Exceeding tolerances of column III.
Center distance.	Basic dimension: +0.017. −0.002.	Basic dimension: +0.018. −0.003.	Exceeding tolerances of column III.
Offset of bridge.	Basic dimension: +.015. −.002.	Basic dimension: +0.017. −0.003.	Exceeding tolerances of column III.
Clearance of bridge with connecting and body loops tangent to a flat surface.	Bridge just touching surface but not scratching same.	Bridge 0.005 below surface.	Exceeding tolerances of column III.
Profile gage.	Slight drag; weight of one additional link sufficient to force through gage.	None.	Slight drag; weight of more than one additional link to force through gage.

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TABLE IA.—*Classification of defects for caliber .30 links*—Continued

Column I Point of inspection—gaging, functioning, and visual	Column II Limits of incidental defects	Column III Limits of minor defects	Column IV Limits of major defects
Extraction strength.	4½ or 10½ lbs.	4 to 11 lbs.	Exceeding limits of column III.
Insufficient neck loop flare to clear shoulder on cartridge case.	Less than column III.	Sufficient interference to prevent proper positioning of link on cartridge case.	Interference with correct positioning of link and scratching of shoulder on cartridge case during linking.
Pull strength.	None.	None.	More than 0.005-inch set on second pull test.
Distorted or nonfunctional links.			(*x*)
Burrs which scratch cartridge cases or feeding components.	Light marks but no shaving of brass.	Definite scratching of cases.	For unusual degree of scratching see next item.
Burrs which affect functioning of the gun or injure personnel in handling.			(*x*)
Laminated links.			(*x*)
Cracks or forming tears which affect functioning.			(*x*)

x indicates that defect is in that category.

NOTE.—Basic dimensions are in inches.

TABLE IB.—*Classification of defects for caliber .50 links*

Column I Point of inspection-gaging, functioning, and visual	Column II Limits of incidental defects	Column III Limits of minor defects	Column IV Limits of major defects
Connecting loop, inside diameter.	Basic dimension: +0.013. −0.013	Basic dimension: +0.015 −0.015	Exceeding tolerances of column III.
Neck loop, inside diameter.	Basic dimension: +0.014. −0.007.	Basic dimension: +0.016. −0.009.	Exceeding tolerances of column III.
Neck loop flare, maximum outside diameter.	Max. dimension: +0.010.	Max. dimension: +0.015.	Exceeding max. tolerance of column III.
Center distance.	Basic dimension: +0.013. −0.018.	Basic dimension: +0.016. −0.021.	Exceeding tolerances of column III.
Offset of bridge.	Basic dimension: +0.022. −0.005.	Basic dimension: +0.027. −0.010.	Exceeding tolerance of column III.
Clearance of bridge with connecting and body loops tangent to a flat surface.	Bridge just touching surface but not scratching same.	Bridge .010 below surface.	Exceeding tolerance of column III.

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TABLE IB.—*Classification of defects for caliber .50 links*—Continued

Column I Point of inspection-gaging, functioning, and visual	Column II Limits of incidental defects	Column III Limits of minor defects	Column IV Limits of major defects
Profile gage.	Slight drag; weight of 1 additional link sufficient to force through gage.	None.	Slight drag weight of more than 1 additional link to force through gage.
Extraction test.	9 to 26 lbs.	8 to 27 lbs.	Exceeding limits of column III.
Insufficient neck loop flare to clear shoulder on cartridge case.	Less than column III.	Sufficient interference to prevent proper positioning of link on cartridge case.	Interference with correct positioning of link and scratching of shoulder on cartridge case during linking.
Pull test.	None.	None.	More than 0.005 inch set on second pull test.
Distorted or nonfunctional links.			(x)
Burrs which scratch cartridge cases or feeding components.	Light marks but no shaving of brass.	Definite scratching of cases.	For unusual degree of scratching, see next inspection item.
Burrs which might affect functioning of the gun or injure personnel in handling.			(x)
Laminated links.			(x)
Cracks or forming tears which might affect functioning or do not pass the pinch test.			(x)

x Indicates that defect is in that category.

NOTE.—Pull test does not apply to caliber .50, M9 link.
NOTE.—Basic dimensions are in inches.

3.3 Hardness.—The links shall be heat treated to the hardness shown on the applicable drawing. (See 4.3.)

3.4 Finish.

3.4.1 *Phosphate coating.*—The links after heat treatment shall be given a phosphate coating conforming to type II, class B of specification 57–0–2. The phosphate coating shall be free of scratches or bare spots.

3.4.2 *Rust preventive.*—The phosphated link shall be coated with a compound rust preventive conforming to grade I of specification AXS–1759. The weight of the residual coating rust preventive compound shall not be in excess of 25 milligrams per link for caliber .50, M2 and M9 links and shall not be in excess of 11 milligrams per link for caliber .30, M1 links.

3.4.3 The phosphated and rust preventive compound coated link shall withstand 24 hours salt spray test as specified in 4.4.1.

3.5 Functioning.—Links shall function properly when assembled into flexible belts. (See 4.5.)

3.6 Workmanship.—Workmanship shall be of a quality conforming to this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—The lot and sublot sizes shall be as specified or authorized by the procuring agency.

4.1.2 *Sampling procedure.*—Inspection samples shall be taken by the Government inspector in accordance with table III or table IV, whichever is used. The total major defectives in the sample shall not exceed the acceptance number

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shown for major defectives. The total number of minor defectives in the sample can exceed the acceptance numbers shown for minor defectives but the combined total of major and minor defectives in the sample shall not exceed the combined total of their acceptance numbers. Incidental defects shall not be used as a basis of acceptance or rejection but shall be cause for corrective action when any single type of incidental defect is repeated more times in a sample than the combined total of the acceptance numbers for minor and major defectives.

TABLE II.—*Normal lot-by-lot acceptance inspection*

Sublot size		500 to 799	800 to 1,299	1,300 to 3,199	3,200 to 7,999	8,000 to 21,999	22,000 to 109,999	110,000 to 549,999	550,000 and over
First sample size		50	75	100	150	200	300	500	1,000
Second sample size		100	150	200	300	400	600	1,000	2,000
Acceptable quality level (percent defective)		Acceptance numbers							
Major	Minor	c_1 c_2	c_1 c_2	c_1 c_2	c_1 c_2	c_1 c_2	c_1 c_2	c_1 c_2	c_1 c_2
0.26–0.50		0 2	1 2	1 3	2 4	2 7	3 8	4 13	7 23
0.51–1.0	0.26–0.50	1 3	2 4	2 6	3 9	4 11	5 13	7 23	14 40
1.1–2.0	0.51–1.0	2 4	3 5	3 8	5 13	6 16	8 25	14 40	← ←
2.1–3.0	1.1–2.0	3 5	4 8	5 11	7 18	9 24	12 25	← ←	← ←
	2.1–3.0	3 9	5 11	6 16	9 23	11 32	16 47	← ←	← ←

←Use samples size in first columns to left in which acceptance numbers are shown for acceptable quality level involved.

Instructions:

1. See 4.1.2.
2. c_1 is the allowable number of defectives in the first sample.
3. c_2 is the allowable number of defectives in the first and second samples combined.
4. If sublot fails to pass on inspection of first sample, take second sample to complete determination of acceptance or rejection of sublot.
5. If 20 consecutive sublots have been accepted in accordance with the above table under conditions of continuous production then table III may be used.
6. In determining the number of defects to be counted against a sample, each link in the sample shall be inspected for both major and minor defects, except that a link discovered to contain a major defect need not be further inspected for minor defects. When more than one defect occurs in the same link only one defect shall be counted against the sample. If one of the defects is classed as major, the major defect shall be counted.
7. Reject each link in which a major or minor defect is found in the above inspections.

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TABLE III.—*Reduced lot-by-lot acceptance inspection*

Sublot size		500 to 799	800 to 1,299	1,300 to 3,199	3,200 to 7,999	8,000 to 21,999	22,000 to 109,999	110,000 to 549,999	550,000 and over
First sample size		10	15	20	30	40	60	100	200
Second sample size		140	210	280	420	560	840	1, 400	2, 800
Acceptable quality level (percent defective)		Acceptance numbers							
Major	Minor	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e	c_1 c_2 c_e
0.26–0.50		0 2 0	0 2 1	0 3 1	1 4 2	1 7 2	1 8 3	1 13 4	2 23 7
0.51–1.0	0.26–0.50	0 3 1	1 4 2	1 6 2	1 9 3	1 11 4	2 13 5	2 23 7	4 40 14
1.1– 2.0	0.51–1.0	1 4 2	1 5 3	1 8 3	2 13 5	2 16 6	3 25 8	4 40 14	← ← ←
2.1– 3.0	1.1 –2.0	1 5 3	1 8 4	2 11 5	2 18 7	3 24 9	4 35 12	← ← ←	← ← ←
	2.1 –3.0	1 9 3	2 11 5	2 16 6	3 23 9	3 32 11	5 47 16	← ← ←	← ← ←

←Use sample size in first columns to left in which acceptance numbers are shown for acceptable quality level involved.

Instructions:

1. See 4.1.2.
2. c_1 is the allowable number of defectives in the first sample.
3. c_2 is the allowable number of defectives in the first and second sample combined.
4. c_e is the allowable number of cumulated defectives in the combined first samples of any group of five consecutive sublots. The first five sublots inspected under table III shall constitute the first group, the second five the next group, and so on, except that whenever a second sample is taken from any sublot, the next sublot shall be the first in a new group of five consecutive sublots.
5. If sublot fails to pass on inspection of first sample, take second sample to complete determination of acceptance or rejection of the sublot.
6. Whenever c_e is exceeded by first-sample defectives in a group of sublots (see instr. 4), a second sample shall be taken on the current sublot. If this sublot then meets the conditions for c_2 the sublot shall be accepted.
7. If any sublot inspected in accordance with the above table fails to pass, inspection under table III shall be discontinued and table II used until the provisions of instruction 5 of table II have again been met.
8. See instructions 6 and 7, table II.

4.2 Inspection.

4.2.1 *Place of inspection and tests.*—Unless otherwise specified, inspection shall be made at the place of manufacture.

4.2.2 *Visual inspection.*

4.2.2.1 Each link in the inspection sample shall be inspected visually by the Government inspector for the defects specified in table IA or table IB, as applicable.

4.2.2.2 A link with laminated metal shall be classed as a defect in accordance with the applicable table.

4.2.2.3 A link having fractures or forming tears extending through the entire stock thickness shall be classed as a functional defect in accordance with the applicable table. Links having slight fractures or forming tears not extending through the stock thickness shall be brought to the attention of the contractor for corrective action but need not be classed as defects unless they are so located on the link as to adversely affect the functioning of the link.

4.2.2.4 Burrs which affect the functioning of the gun, scratch the cartridge case or feeding components, or injure personnel in handling are not acceptable and shall be cause for rejection. Each such burred link shall be classed as a defect in accordance with the applicable table.

4.2.3 *Strength inspection.*

4.2.3.1 *Pull strength.*

4.2.3.1.1 After phosphating and coating with rust preventive compound, the contractor shall subject each link to the required pull. The links shall be assembled on two parallel studs of such diameters as to fit easily into the loops of the link. The studs shall then be forced apart laterally to produce the pull specified on the applicable drawing.

4.2.3.1.2 The government inspector shall subject 5 percent of the inspection sample but not less than 10 links to the load required by the applicable drawing. If one link shows more than 0.005 inch increase in center distance, the entire sample shall be subjected to the load

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required. Each link showing more than 0.005 inch increase in center distance as a result of this repetition of the contractor's pull inspection shall be classed as a defect in accordance with the applicable table.

4.2.3.2 *Extraction strength.*—Links shall be inspected for required extraction strength by inserting a smooth steel plug into the body and neck loops and then determining the force in pounds required to extract the plug. The plug shall be made to the minimum dimensions of the applicable cartridge case. The force required to extract the plug shall be within the limits shown on the applicable drawing. Twenty percent of the acceptance sample but not less than 50 links shall be gaged for extraction effort. If one link is found to be outside the allowable range, the entire sample shall be gaged. Each link failing to meet the inspection shall be classed as a defect in accordance with the applicable table.

4.2.4 Dimensional inspection.

4.2.4.1 Each link in the inspection sample shall be inspected for the dimensional defects listed on the applicable table.

4.2.4.2 Gaging for dimensions shall follow contractor's pull inspection, but shall precede any further inspection which could alter the link dimensions (extraction or inspector's pull inspections).

4.2.4.3 In gaging loop diameters, the links shall be considered acceptable if they pivot freely about or drop off the "GO" portions and fail to pivot freely or fail to drop off the "NO-GO" portions of the required gage members.

4.2.4.4 The tendency of a link to pivot about, fall through, or drop out of a gage of its own weight shall be used as the standard for determining conformity with the gage.

4.3 Hardness test.—Prior to rustproofing and after heat treating, the inspector shall check links at random for hardness to insure compliance with the applicable drawing. The number of links tested shall be based upon production quantities and previous quality but shall never be less than 3 links per hour of operation for each heat treating unit. In the event that the hardness of the links is not within the required limits, the links in process shall be rejected, subject to removal by the contractor of all links not meeting the requirements.

4.4 Finish test.

4.4.1 The salt spray test shall be conducted as specified in specification QQ–M–151. The links shall be taken at random after they have been coated with compound rust preventive. At least five links for each 4 hours of the phosphate coating process shall be taken for testing. The selected links shall be placed in a new, unused link carton for a period of 1 hour to absorb any surplus compound rust preventive and then subjected to the salt spray test. The results of this test shall not be used as a basis of acceptance or rejection of the links already processed, but shall be called to the attention of the contractor. After any unsatisfactory test the links thereafter processed shall be subject to rejection and final acceptance, but shall be withheld until satisfactory salt spray tests have been completed. After a series of five consecutive lots have successfully met the salt spray test, the original procedure shall again be resumed. When a majority of the links in any test show any signs of corrosion except at the sharp edges, at the completion of the test, the sublot shall be considered a failure. When one or two links show not more than two areas of corrosion and each such area does not exceed ⅛ of an inch in diameter or its equivalent, the test shall be acceptable unless such test is a repetition of a similar result in any 24-hour period, in which case it shall be a failure. Links having scratches or bare spots in the phosphate coating shall not be selected for salt spray testing but shall be brought to the attention of the contractor for corrective action; such links shall not be cause for rejection except as provided in 4.6.2.

4.4.2 Evidence of rust preventive compound in excess of the requirements of 3.4.2 shall be checked by a weight method. Except as provided in 4.6.2, the results of this test shall not be used as a basis for acceptance or rejection of the links already processed but shall be called to the attention of the contractor for corrective action on links thereafter processed.

4.5 Functioning test.—On initial contracts 100 links taken at random from the first lot

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shall be forwarded to Rock Island Arsenal, Rock Island, Illinois, for functioning test and approval prior to quantity production.

4.6 Rejections.

4.6.1 When a sublot of links submitted by the contractor has more than the allowable number of defectives in the inspection sample, the entire sub-lot shall be rejected for reinspection and removal of all defective links before resubmission to the Government inspector. Failure of a sublot on resubmission shall be cause for final rejection of the entire sublot.

4.6.2 Items of inspection specified in 4.1.2, 4.2.2, 4.4.1 and 4.4.2 not normally used as a basis for acceptance or rejection shall be subject to corrective action. Failure on the part of the contractor to take corrective action at a time satisfactory to the Government inspector shall be sufficient cause for the Government inspector to reject all sublots containing such defective link processed thereafter.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Packaging and packing shall conform to applicable drawings and specifications.

5.2 Marking.—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with specification 100–2; for the Navy, in accordance with the Navy Shipment Marking Handbook; and for the Air Force in accordance with specification 94–40645.

6. NOTES

6.1. Ordering data.—Purchase instrument should specify the following:

a. Title, number and date of specification.

b. Type required. (See 1.2.1.)

6.2 Steel covered by specification 57–136–1 has been used successfully for the manufacture of these links, but the contractor is not obligated to use steel of this specification.

6.3 Corrective action with reference to slight fractures or forming tears may not result in their complete elimination but the contractor is required to exert constant effort in this direction.

6.4 The Government inspector should carefully check center distance before and after the pull inspection to determine the amount of change produced by the inspection. The inspection cannot be rechecked on the same link since the change will become less on each successive inspection. An increase in center distance over 0.005 inch on the ordnance inspection indicates that the contractor is not putting the proper pull on the links, that the heat-treatment is improper, or that certain physical dimensions are undersize, resulting in a weak link.

6.5 The classification of defects as incidental, minor, or major will be determined by the degree of defect rather than by the nature of the defect. See table I for tabulation of defects.

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